



Baryte

△ Baryte rough on sphalerite

The name “baryte” originates from the Greek *barys*, meaning “heavy”, a reference to the mineral’s high specific gravity. It is very soft and breaks readily in a number of directions – it is faceted with difficulty, purely as a collector’s gem. Golden baryte from Colorado, USA, is the most prized gemstone colour; blue baryte is also faceted for collectors. Baryte is found in the form of stalactites, too, and round, banded sections of these are sometimes polished and mounted in silver frames as pendants. It is the most important single source of barium (see box, below).

Specification

Chemical name Barium sulphate | **Formula** BaSO₄
Colours Colourless, golden, bluish, greenish, beige | **Structure** Orthorhombic | **Hardness** 3–3.5 | **SG** 4.5 | **RI** 1.63–1.65
Lustre Vitreous, resinous, pearly | **Streak** White | **Locations** England, Italy, Czech Republic, Germany, Romania, USA

Golden colouring



Prismatic crystals | Rough | This group of baryte crystals exhibits a prismatic form and the resinous lustre characteristic of the mineral.

Tabular crystals



Golden baryte crystals | Rough | The golden baryte material that is mined near Canyon City, Colorado, USA, is world-famous for its crystal forms and colours, as can be seen in this brightly coloured specimen.

Flake-like form



Baryte crystals | Rough | Originating from Wet Grooves Mine, Yorkshire, UK, this specimen consists of a large group of tabular baryte crystals.

Medicine and industry

Baryte and barium

Baryte powder is used in medicine as a “barium meal” for imaging the stomach and intestine. It is also an important mineral in oil and gas production, used as drilling mud to seal boreholes and prevent oil or gas blowout in oil- and gas-wells, which accounts for about 70 per cent of its industrial output. It is also used as a filler in paper- and cloth-making, and as an inert body in coloured paints.



Barium meal Barium sulphate is visible in X-rays when ingested by the patient.



Baryte crystal | Rough | This tabular, double-ended crystal shows some damage on its left side. Growth zones can be seen in pale banded sections at its base.

Rectangular table facet



Mixed cut gemstone | Cut | Baryte is one of the most difficult of all collector’s gems to facet. The cutter has done a fantastic job with this stone.